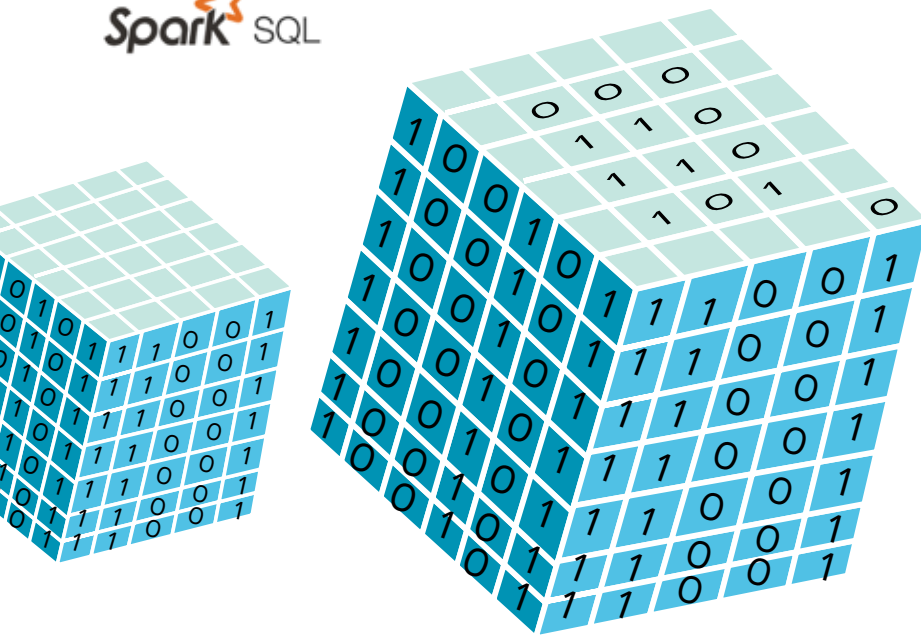


Structured Data Processing with Spark SQL

Spark SQL is the module in the Spark ecosystem that processes data in a structured format. It internally uses the Spark Core API for its process, but the usage is abstracted from the user. This article dives a little deeper and tells you what's new in Spark SQL 3.x.



and data sources (relational and non-relational). Spark SQL supports a unified DataFrame interface to process different types of sources, as given below.

Files

- CSV
- Text
- JSON
- XML

JDBC/ODBC

- MySQL
- Oracle
- Postgres

Files with schema

- AVRO
- Parquet

Hive tables

- Spark SQL also supports reading and writing data stored in Apache Hive.

With DataFrame, users can seamlessly read these diversified data sources and do transformations/joins on them.

What's new in Spark SQL 3.x

In the previous releases (Spark 2.x), the query plans were based on heuristics rules and cost estimation. The process from parsing to logical and physical query planning, and finally to optimisation was sequential. These releases had little visibility into the runtime characteristics of transformations and actions. Hence, the query plan was suboptimal because of

With Spark SQL, users can also write SQL-styled queries. This is essentially helpful for the wide user community that is proficient in structured query language or SQL. Users will also be able to write interactive and *ad hoc* queries on the structured data. Spark SQL bridges the gap between resilient distributed data sets (RDDs) and relational tables. An RDD is the fundamental data structure of Spark. It stores data as distributed objects across a cluster of nodes suitable for parallel processing. RDDs are good

for low-level processing, but are difficult to debug during runtime and programmers cannot automatically infer schema. Also, there is no built-in optimisation for RDDs. Spark SQL provides the DataFrames and data sets to address these issues.

Spark SQL can use the existing Hive metastore, SerDes, and UDFs. It can connect to existing BI tools using JDBC/ODBC.

Data sources

Big Data processing often needs the ability to process different file types